Analysis and Evaluation of Hardware Trust Anchors in the Automotive Domain

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ABSTRACT

Automotive architectures get increasingly more complex both regarding internal as well as external connections to offer new services like autonomous driving. This development further broadens the cyberattack surface of modern vehicles. As mitigation mechanism, hardware trust anchors (HTAs) are increasingly integrated into the electronic control units (ECUs) of modern vehicles to shield security-sensitive data like cryptographic keys against a variety of cyberattacks. However, the provided security capabilities differ among the HTAs. There is currently no evaluation of the HTAs that also addresses current and emerging future requirements of the automotive domain. Thus, in this work, we will analyze and evaluate typical automotive HTAs regarding their feasibility to be used in modern and upcoming vehicle architectures. For this we derive comprehensive evaluation criteria from both related work as well as the automotive domain analysis and make an extensive assessment of the HTA properties in accordance to requirements of the automotive domain.

CCS CONCEPTS

• **Security and privacy** → **Domain-specific security and privacy architectures**; **Trusted computing**; *Embedded systems security*; *Hardware-based security protocols.*

KEYWORDS

automotive cybersecurity, hardware trust anchor (HTA), evaluation, Trusted Platform Module (TPM), Hardware Security Module (HSM), EVITA, Secure Hardware Extension (SHE), ARM TrustZone

ACM Reference Format:
Christian Plappert, Andreas Fuchs, and Ronald Heddergott. 2022. Analysis and Evaluation of Hardware Trust Anchors in the Automotive Domain. In *The 17th International Conference on Availability, Reliability and Security (ARES 2022), August 23–26, 2022, Vienna, Austria.* ACM, New York, NY, USA, 11 pages. https://doi.org/10.1145/3538969.3538995

1 INTRODUCTION

Recent developments in the automotive industry show that automotive Electrical / Electronic (E/E) architectures are becoming increasingly complex to meet the ever-growing demand for new services like modern Advanced Driver-Assistance Systems (ADAS). As of today, the software in architectures of modern vehicles comprises more than 100 million Lines of Code (LOC) distributed among up to 100 heterogeneous computing units, so-called Electronic Control Units (ECUs). For comparison, this massive code size surpasses for example Facebook and or whole operating systems like Windows 7 or Debian 5.0 [20]. In particular with the rise of fully autonomous vehicles, this trend towards complexity will further increase and will not only comprise the internal but also external communications of the vehicle in order to support modern vehicle services.

The increase in complexity is also reflected in future vehicle E/E architecture development. In contrast to current domain-based architectures, the performance required to support these new functions will be concentrated in a few central High-Performance Computers (HPCs) that are then connected to legacy ECU networks [7].

With the increasing complexity both with regard to the internal as well as external interfaces of modern vehicles, also their attack surface broadens. Recent attacks show that the vehicle is vulnerable to a variety of different attacks ranging from stealing vehicles [22] to the hijacking and remote controlling of whole vehicles [11]. These attacks show that the interfaces are highly critical and typically the first step in an attacker attack chain [24].

Current state-of-the-art approaches utilize so-called Hardware Trust Anchors (HTAs) to mitigate against a variety of cyberattacks. The HTAs use hardware isolation mechanisms to shield security-sensitive data, e.g., cryptographic keys, and operations, e.g., signature generation, from the possibly compromised host and thus mitigate against both software and also even hardware attacks. In the automotive domain HTAs are currently just started to be increasingly used in ECUs across the whole vehicle [27].

There are different standardized feature sets for HTAs which are prominent in the automotive domain. These are the Secure Hardware Extension (SHE) and the EVITA Hardware Security Module (HSM). HSM is a term coined by the European research project EVITA [4], which is now used to describe all kinds of different hardware security subsystems that enable isolation. Recently, also the Trusted Platform Module (TPM) that is already widespread in consumer electronics is slowly started to be integrated to secure modern vehicle architectures [15].

Different hardware manufacturers use different names for their HTAs. For example, Infineon offers SHE+ for their first and EVITA drivers for their second generation Aurix [14], Renesas provides SHE/EVITA drivers for their Intelligent Cryptographic Unit (ICU) HSM [26], and NXP Semiconductors has with the Hardware Security Engine (HSE) an HSM compliant with SHE and EVITA [32].

From this short overview, it is already clear that there is a huge variety of HTA technologies to secure automotive applications. Each

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ARES 2022, August 23–26, 2022, Vienna, Austria
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ACM ISBN 978-1-4503-9670-7/22/08.
https://doi.org/10.1145/3538969.3538995

of these technologies have their corresponding advantages and disadvantages with respect to different evaluation criteria. While EVITA and SHE are still the predominant HTAs used in the automotive domain, they are quite dated and may not be appropriate to secure upcoming vehicle trends. However, the Trusted Platform Module (TPM) is broadly integrated into consumer electronics, but just started to be used in the automotive context and must first be analyzed for its feasibility.

Thus, in this work we analyze established as well as promising HTAs for securing automotive applications and evaluate them regarding relevant evaluation criteria that we derive from literature as well as typical automotive use cases to secure current and future automotive E/E architectures.

In short, we make the following contributions:

(1) We propose a new HTA evaluation approach that integrates requirements from related work as well as results from an automotive domain analysis.
(2) We select both well-established as well as new promising HTAs for securing current and emerging automotive architectures.
(3) We conduct an extensive automotive domain analysis in which we define a future automotive E/E reference architecture. From this we derive threat and attacker models as well as relevant automotive use cases.
(4) We conduct the HTA analysis based on the corresponding specifications and evaluate them regarding our derived evaluation criteria that we derived both from related work as well as our automotive domain analysis.

To the best of our knowledge, there is currently no similar recent work addressing the analysis and evaluation of automotive HTAs. Especially not with regard to the requirements of the automotive domain and an extensive analysis of corresponding specifications.

The remainder of the paper is structured as follows: Section 2 gives an introduction into the automotive domain and the later analyzed HTAs. Moreover, we reference similar related work. In Section 3 we conduct our automotive domain analysis where we derive a possibly future E/E reference architecture, define our attack and threat model, and derive relevant security-critical automotive use cases. In Section 4 we derive our evaluation criteria based on the related work and our automotive domain analysis. In Section 5 we carry out our analysis and evaluate the different HTAs based on the derived criteria. Finally, we summarize our results Section 6 and conclude our work in Section 7.

2 BACKGROUND AND RELATED WORK

This section gives some background on the automotive domain and analyzed technologies. Moreover, it presents related work.

2.1 Automotive Domain

Automotive Electrical / Electronic (E/E) architectures are constantly evolving and the increasing demand of vehicle services is reflected by the various design principles by which their topologies are clustered [7]. Early architectures are dominated by a central gateway that interconnects several Controller Area Network (CAN) subnetworks with corresponding Electronic Control Units (ECUs). However, with the characteristic of CAN as shared medium bandwidth became a bottleneck for new features. Thus, modern vehicles are typically designed with domain-based architectures that are for example shown in [24]. With the emerge of autonomous vehicles, the network topology will further change to respect new performance requirements of the autonomous driving task. In Section 3, we define a possibly future architecture for autonomous vehicles that we base on [7].

2.2 Hardware Trust Anchor Technologies

State of the Art HTA technologies can be clustered into (1) dedicated security chips, (2) security subsystems, and (3) Trusted Execution Environments (TEEs).

A dedicated security chip is a standalone silicon that communicates via distinct interface with other components. Examples for dedicated security chips are TPMs or Smartcards. Security subsystems are isolated regions on the host system that execute security operations. Examples are the Secure Hardware Extension (SHE) and Hardware Security Modules (HSMs). Lastly, TEEs instantiate a secure environment where secure applications can be deployed utilizing the peripherals and modules of the host. A widespread TEE is for example ARM TrustZone [18]. HSM and SHE are currently used as state-of-the-art technologies in the automotive domain, while the TPM is just started to be integrated [15]. Since more and more vehicle controllers are based on ARM architectures [14], also TrustZone becomes a promising HTA to secure future architectures.

2.2.1 Trusted Platform Module 2.0. The Trusted Platform Module 2.0 (TPM 2.0)[1] and its corresponding software stack are openly specified by the Trusted Computing Group (TCG). The TPM is a security co-processor featuring some advanced capabilities going beyond typical key management and usage functionality.

The TPM features a True Random Number Generator (TRNG), key generators, engines for (a-)symmetric cryptography and hash operations, and (non-) volatile tamper-proof memory. The non-volatile memory may be used to persist keys, monotonic counters, and arbitrary data across boot cycles and may secure these regions against access by third parties if necessary. To do so it has some advanced access protection mechanisms reaching from simple password-based authorizations to session-based authorizations using cryptographic keys or complex access policies. The volatile memory can store current used keys, session information and contains some exclusive registers for the storage of integrity measurements, the Platform Configuration Registers (PCRs). The PCRs can only be extended by adding a new measurement value to the already stored one and thus building a hash chain over all measurements before reset. The measured values in the PCRs can be used to unlock keys only in a benign software state (sealing) or securely report the platform state to an external entity (remote attestation) to establish a trust relationship.

2.2.2 Secure Hardware Extension (SHE). SHE is an on-chip security extension for Microcontroller Units (MCUs). It is specified by the German Original Equipment Manufacturer (OEM) software initiative (HIS, german for Herstellerinitiative Software), a committee drawn from Audi AG, BMW AG, Daimler AG, Porsche AG,

[1] For brevity in the following simply referred to as TPM.

and Volkswagen AG with the goal to define harmonized interfaces (product and process) to reduce supplier effort to adapt to differing OEM requirements.

The general goal of SHE is to isolate cryptographic keys from the rest of the controller in a hardware-backed shielded location and thus to secure them against various attacks. SHE is not intended to replace high security solutions like TPMs or smartcards since the specification, among other things, does not require SHE to be resilient against physical attacks. SHE consists of 3 components: a memory to store cryptographic keys and their meta data, cryptographic engines, e.g., to implement block ciphers like AES, and control logic that connects SHE to the Central Processing Unit (CPU) of the controller. SHE can be implemented as state machine or as part of a small, dedicated CPU core. Manufacturer specific enhancements of the basic specification are labeled as SHE+.

2.2.3 EVITA Hardware Security Module (HSM). Within the research project EVITA [4], which was carried out by representatives from automotive industry and academic research partners, three different variants of HSMs were specified for the application in the automotive environment [5]: EVITA *full*, *medium* and *light* HSMs that offer different capabilities for various application scenarios. While EVITA *full* is intended to be used in Vehicle-to-Everything (V2X) scenarios where strong performance requirements for signature generation and verification have to be met, EVITA *medium* is intended to be used for in-vehicle ECU communication, and EVITA *light* is used for sensor/actuator communication. Besides a shielded location, the EVITA HSM features ECU Configuration Registers (ECRs) that have a similar function to the TPM PCRs and enable trusted computing concepts like sealing and remote attestation.

2.2.4 ARM TrustZone. ARM licenses processor designs and divides there newer Cortex processor product range in three groups based on the application spaces: high-end application processors (Cortex-A), high-performance real-time processors (Cortex-R) and resource-constrained microcontroller processors (Cortex-M).

Some of the Cortex-A and Cortex-M processor types can support the ARM TrustZone security extension that is a low-cost alternative to adding an dedicated security core to a System on Chip (SOC). Processors with the TrustZone security extension use a hardware-backed access control to switch between two states: the non-secure and secure world. The switching mechanism is orthogonal to all other capabilities of the processor and each core can operate independently of the user while using the same core. Memory and peripherals are made aware of the current operating world. The non-secure world is meant to execute normal applications and the secure world is meant to execute security-sensitive software components (secure boot, cryptographic libraries, authentication) and security resources (e.g., secure storage, cryptographic accelerators, random number generators). Non-secure software can only access memory and peripherals of the non-secure world while secure software can access resources in both worlds.

2.3 Related Work

There is some related work regarding the evaluation of HTAs for the automotive domain. In [36], the authors propose a vehicular HSM for the protection of in-vehicle ECUs and their communications. Next to a description of hardware design and details on a prototypical implementation as well as performance evaluation, the authors provide a comparison of their HSM to other already existing HTA solutions. In [13], the authors evaluate secure communication approaches for in-vehicle networks and among others also give some background to SHE, the EVITA HSM and the TPM. For the HSM of an NXP MPC5748G, they also report some performance measurements of AES-128 encryption. In [37] different in-vehicle key management systems are analyzed and they make a short high-level comparison of the EVITA HSM and the TPM regarding costs, provided security, and functionality. A whitepaper by NXP [30] describes concepts and usage paradigms of security subsystems. The define a security subsystem as central or distributed component integrated within a larger Integrated Circuit (IC).

The analysis of related work shows that a comprehensive analysis of automotive HTAs is quite rare. Existing work is either quite old [36] and does not consider current developments, analyze the topic from a more generic level [13, 37], or focus on one specific HTA [30]. However, we use the contents of related work to derive some of our evaluation criteria and if necessary update their evaluation results with current developments.

3 AUTOMOTIVE DOMAIN ANALYSIS

This section is dedicated to the analysis of properties and requirements of the automotive domain. Thus, we define a modern reference architecture, threat model, and typical automotive use cases.

3.1 Automotive Reference Architecture

The increasing development towards automation also necessitates an increase in the performance of the in-vehicle network architectures. Based on the design principles introduced in Section 2 and current developments in the automotive domain, we derive a future centralized E/E architecture that is depicted in Figure 1.

This architecture is characterized by central High-Performance Computers (HPCs) that are interconnected via a high-performance network, e.g., Automotive Ethernet. These HPCs are both utilized to compensate the growing performance demand introduced by the development towards automated driving as well as to meet safety requirements introduced by the automation. The HPCs are also connected to legacy subnetworks that comprise of the typical resource-constrained ECUs and automotive bus systems, e.g., CAN.

3.2 Automotive Threat and Attacker Model

As depicted in Figure 1, modern vehicles are strongly interconnected to external entities via various interfaces. These provide a wide attack surface for remote cyberattacks either directly from backend systems that are connected via cellular networks or relayed from other external parties that are themselves connected to the internet, e.g., charging stations, repair shops, or mobile devices. Additionally, unlike backend server systems, vehicles and their internal network are directly exposed to physical access of attackers.

From this we define three attacker types with increasing capabilities but, as our previous work has shown [24], with decreasing attack feasibility: The communication channel attacker (Type1), the hijacking attacker (Type2), and the physical attacker (Type3).

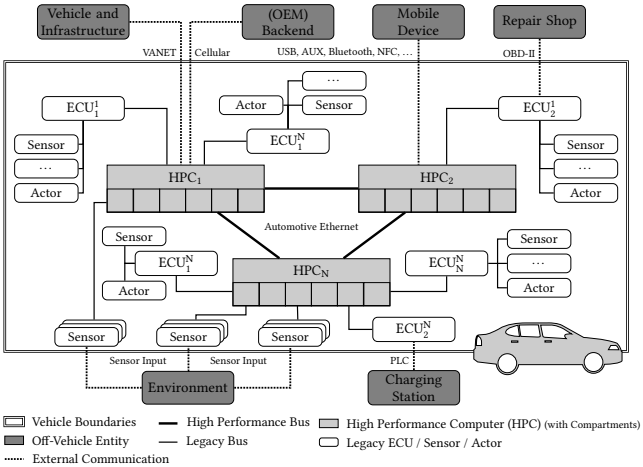


Figure 1: Future Automotive E/E Reference Architecture

The Type1 attacker is the classic Dolev-Yao attacker [8] that can access all external and internal communication channels of the vehicle and inject, replay, modify, relay, and/or intercept transmitted messages. However, he cannot break any cryptographic algorithms.

The Type2 attacker is additionally able to hijack ECUs and execute privilege escalation attacks to circumvent software security mechanisms like Mandatory or Discretionary Access Control.

The Type3 attacker is able to modify underlying hardware mechanisms, e.g., read, modify, or delete security-sensitive memory, physically manipulate or deactivate critical IC functionality like sensors, or execute various sidechannel attacks, e.g., simple or differential power analysis or electromagnetic analysis. This attacker model corresponds to the cyberphysical Dolev-Yao attacker [28].

We will use this attacker model later on, to evaluate the security guarantees of the analyzed HTAs.

3.3 Automotive Use Cases

In [27] current trends regarding cybersecurity mechanisms are detailed. By 2023, all mechanisms that are now seldom used, will be common to many vehicle ECUs. These mechanisms are for example the broad usage of Public Key Infrastructures (PKIs), HTAs, signed software updates, secure and authenticated boot, and secure external and internal communications.

For these mechanisms we define now the corresponding use cases, which we will later use to map our HTA evaluation criteria. The use cases are UC1: Credential Management, UC2: ECU Communication, and UC3: HPC Utilization.

3.3.1 UC1: Credential Management. Vehicles are assembled from ECUs made by different manufacturers. Partially, the ECUs are already initialized with credentials, e.g., cryptographic keys, certificates, and/or unique identifiers like serial numbers, of the corresponding vendors. If not already done, the ECUs need to be initialized with the credentials, e.g., with key creation or derivation, and the credentials need to be securely exchanged among relevant in-vehicle ECUs or other parties in the vehicle environment, e.g., backend systems. This enables mutual trust relationships and is the prerequisite for all use cases security concepts.

With typical life cycles of vehicles with over 20 years [5], it is also crucial to be able to securely update already distributed credentials, e.g., in case keys were leaked, certificates expired, or cryptographic primitives and schemes are broken, to mitigate against the upcoming vulnerabilities.

3.3.2 UC2: ECU Communication. Modern connected vehicles exchange a plethora of messages within the in-vehicle network as well as to external entities. To support ADAS these messages also contain control signals to maneuver the vehicle. In order to guarantee safety requirements, trust in these messages need to be established.

This is typically done by using cryptographic schemes. However, due to bandwidth limitations in the in-vehicle networks, this is done only for the messages of major ECUs and then mostly limited to using, e.g., Automotive Open System Architecture (AUTOSAR)'s Secure Onboard Communication (SECOC) that utilizes symmetric Message Authentication Codes (MACs) to enable integrity and authenticity of the exchanged messages.

With future network architectures, also network and processing capacities of the ECUs increase, so that also schemes with easier key management like asymmetric cryptography to enable Transport Layer Security (TLS) or encryption may be implemented to enable new services.

3.3.3 UC3: HPC Utilization. This use case takes into account the trend towards full automation and the corresponding application scenarios and necessary developments in automotive E/E architectures. It is especially targeted at upcoming challenges regarding the secure consolidation of Virtual ECUs (vECUs) on one platform and corresponding redundancy and safety concepts.

4 EVALUATION CRITERIA DEFINITION

From our automotive domain analysis as well as from related work, we derived the evaluation criteria EC1: Security Services, EC2: Platform Security, EC3: Flexibility, EC4: Performance, EC5: Software Ecosystem, and EC6: Cost Efficiency that we will detail in the following.

4.1 EC1: Security Services

Security services provide the actual functionality to the host environment. They can be used to ensure the confidentiality, integrity and authenticity of data. We cluster them into (1) auxiliary, (2) basic, and (3) advanced services.

4.1.1 Auxiliary Security Services. Auxiliary security services help to provide basic or advanced services and to implement higher-level security concepts. These services rely on hardware features and thus cannot be emulated using software measures with the same security guarantees.

Non-Volatile Memory (NVM): Non-volatile memory can used to store arbitrary information as well as specific access protected data structures like monotonic counters or bitfields. This service can be used to store public information like certificates or public keys integrity-protected or define obligations based on status of counters or bitfields.

Secure Internal Clock (SIC): This service enables a hardware-protected time reference that may be synchronized with UTC time.

Random Number Generator (RNG): Strong Random Number Generators (RNGs) are essential for many cryptographic operations, like nonce generation or secret key generation. Typically, Pseudo Random Number Generators (PRNGs) are seeded internally from a TRNGs or externally from a TRNG during production.

4.1.2 Basic Security Services. These services provide basic security functionality from cryptographic engines to key management.

Asymmetric Cryptographic Engines (ACE): This service enables the creation and verification of digital signatures as well as data en- or decryption using asymmetric key pairs.

Symmetric Cryptographic Engines (SCE): This service enables the creation and verification of symmetric authenticators, like MACs. Also, it enables secret key-based data encryption and decryption.

Cryptographic Hash Functions (CHF): This service enables the generation and verification of hash fingerprints and Hash-based Message Authentication Codes (HMACs). The later are generated with a secret key.

Key Usage: With this service key usage can be bound to various obligations. This could be for example usage restrictions, like digital signature generation and verification, MAC generation and verification, encryption and decryption, or usage authorization mechanisms like simple password protection, session-based authorization, or platform sealing.

Key Management: This service allows to securely manage cryptographic keys. This includes key creation, key agreement (e.g., Diffie-Hellman), key derivation as well as key import from and export to trusted external parties.

4.1.3 Advanced Security Services. These services go beyond traditional key management and usage functionalities. They are relevant for more advanced use cases.

Secure and Measured Boot: This service allows to ensure the integrity of the system. In case of secure boot this is done by verifying the authenticity of software images, e.g., by verifying digital signatures, before executing. If the verification is unsuccessful the bootprocess is typically stopped. In case of measured boot, the going-to-be executed code is measured (hashed) before handing-over control. The measurements are securely stored in a Root of Trust for Storage (RoTS), e.g., TPM, and can be reported to external entities. Secure and measured boot are orthogonal security mechanisms, where secure boot ensures the execution of a benign software, and measured boot ensures the execution of the correct software.

Attestation: This service allows to prove the trustworthiness of a system, e.g., its integrity, to a third party party. Typically the measured values obtained from a measured boot are authenticated and send to the remote verifier.

Audit: The audit service allows an (external) auditor to trace that operations where successfully executed within the HTA.

Hardware Watchdog Timer: A Hardware Watchdog Timer (HWDT) is a privileged component that may trigger safety mitigations of the subcomponent, e.g., failsafe modes, in case a periodically triggered challenge-response mechanism fails.

4.2 EC2: Platform Security

The provision of a shielded location is the main advantage of all HTA technologies. The shielded location protects security-sensitive data from unauthorized access and depending on the specific HTA type also against physical tampering.

The platform security evaluation criterion evaluates the security guarantees of the provided shielded location of the analyzed HTAs against our defined attacker model. Taken into account in the platform security evaluation are the tamper-resistance properties against physical attackers, the verifiability of the security properties, and an analysis of the provided Trusted Computing Base (TCB). The TCB is a minimal set of hardware and software components that must be trusted as uncompromised in order to fulfill the system security guarantees.

4.3 EC3: Flexibility

This evaluation criterion evaluates the usage flexibility of the corresponding HTA. This both addresses the flexibility to adapt the HTA to necessary features to support the use cases as well as the crypto-agility to mitigate against upcoming attacks. The latter is in particular important due to long lifecycle of vehicles in the field with over 20 years [5].

4.4 EC4: Performance

Certain automotive use cases, e.g., V2X message exchanges [1], require a high-performance throughput of cryptographic operations. This evaluation criteria compares the relative performance of the analyzed HTAs regarding processing speed or possibly available cryptographic hardware accelerators.

4.5 EC5: Software Ecosystem

This criterion analyzes the available software for a HTA technology. This includes the firmware, driver, as well as software stack and application software. Also, the software licenses and the potential associated additional costs are analyzed.

4.6 EC6: Cost Efficiency

The cost efficiency criterion analyzes the additional costs that come with the application of the corresponding hardware trust anchor technology. The costs depend on two factors (1) the cost of the software and (2) the readiness of the software. Thus, this criterion partially overlaps with EC5.

5 HARDWARE TRUST ANCHOR EVALUATION

In this section the different HTAs are evaluated with respect to our defined evaluation criteria as well as the aspects from our automotive domain analysis. We base the analysis on the corresponding HTA specifications for SHE [9], EVITA [5, 36], and TPM [35].

Table 2 shows which evaluation criteria is influenced by which aspects of our automotive domain analysis for our assessment.

We will detail about each criteria in the following and summarize our findings in Section 6.

Table 1: Security Services Comparison

Technology	Auxiliary Services			Basic Services					Advanced Services
	NVM	SIC	RNG	ACE	SCE	CHF	KU	KMA	
SHE	keyslot	–	PRNG (TRNG seed)	–	AES-128 ECB/CBC	CMAC	enc, MAC	export (RAM key only), import	Secure Boot
EVITA *light*	(arbitrary, counter)[1]	–	AES-PRNG (ext. Seed)	–	AES-128 CCM/GCM	–	enc, auth, derivation, DH, auth (PW, integrity)	creation, derivation, exchange, import, export	(Secure & Measured Boot)[1] [Auditing, Attestation][2]
EVITA *medium*	arbitrary, counter	UTC clock	AES-PRNG (TRNG seed)	–	AES-128 CCM/GCM	–	cf. EVITA *light*	cf. EVITA *light*	Secure & Measured Boot [, Auditing, Attestation][2]
EVITA *full*	arbitrary, counter	UTC clock	AES-PRNG (TRNG seed)	ECC-256	AES-128 CCM/GCM	Whirlpool, HMAC	cf. EVITA *light*	cf. EVITA *light*	Secure & Measured Boot [, Auditing, Attestation][2]
TPM 2.0	arbitrary, monotonic counter, bitfield, extend, pin	UTC clock	TRNG	RSA, ECC,	(AES)	HMAC, CMAC	enc, auth, derivation, DH, transport, auth (pw, Session (key, policy))	creation, derivation, exchange, import, export, duplication, certify creation	Measured Boot, Attestation (ROTR, Quote Command), Audit Sessions, HWDT
HSM (no FW)	–[3]	–[3]	–[3]	FP	FP	FP	FP	FP	FP
TrustZone	–[3]	–[3]	–[3]	FP	FP	FP	FP	FP	FP

[1] Optional, [2] Soft Requirement, [3] HW requirement and cannot be realized by software, ACE: Asymmetric Cryptographic Engine, CHF: Cryptographic Hash Function, FW: Firmware, FP: Freely Programmable, HWDT: Hardware Watchdog Timer, KMA: Key Management, KU: Key Usage, NVM: Non-Volatile memory, RNG: Random Number Generator, SCE: Symmetric Cryptographic Engine, SIC: Secure Internal Clock

Table 2: Mapping Evaluation Criteria to Automotive Domain Analysis

	EC1	EC2	EC3	EC4	EC5	EC6
Use Case	●		●	●		
Attacker Model	●	●	●			
Automotive Domain					●	●

5.1 Security Services

Table 1 shows an overview of the security services provided by the different HTAs. The HTAs are roughly ordered by the variety of the offered security services. SHE and EVITA *light* are intended as lightweight HTAs and thus offer only a subset of the security services, while the HTAs without fixed feature set, in particular the generic HSM without firmware as well as TrustZone, offer at least all services that can be realized by software since they are freely programmable. Hardware-dependent features like Non-Volatile memory (NVM), Secure Internal Clock (SIC), TRNG, and a HWDT thus depend on the specific implementation.

From the HTAs with a fixed feature set, both SHE and EVITA *light* are designed for lightweight use cases and thus offer the most limited set of security services. The major limitations come from the lack of asymmetric cryptography and of a SIC. At least SHE features a TRNG that seeds its PRNG, NVM that can however only be used as keyslot and to store the MAC for secure boot, and Cipher-based Message Authentication Code (CMAC) as Cryptographic Hash Function (CHF). However, EVITA *light* has only optional NVM and secure/measured boot, a PRNG that needs to be externally seeded by a TRNG, but supports more secure Advanced Encryption Standard (AES) encryption modes and more fine granular key usage including key authorizations.

EVITA *medium* and EVITA *full* build up on the EVITA *light* feature set and add mandatory NVM as well as boot integrity protection with secure and measured boot, a SIC, WHIRLPOOL and HMAC as cryptographic hash functions, and a TRNG as seed for the PRNG. On top, EVITA *full* also implements an asymmetric cryptographic engine and supports Elliptic Curve Cryptography (ECC). However, for the advanced features like auditing or attestation, EVITA specifies only soft requirements that a EVITA HSM should be able to provide without more details about the specific implementation.

Finally, the TPM offers the most comprehensive set of security services. It is mostly compliant to the security services of the other HTAs and enhances or adds features. It has a very flexible NVM that can store arbitrary data as well as specified data types, like monotonic counters, bitfields, dedicated "extend" fields to store integrity measurements, and PIN indices to protect against dictionary attacks. Further, it can be utilized for secure and measured boot and acts as Root of Trust for Reporting (ROTR) to enable trustworthy attestation mechanisms. Dedicated audit sessions can be utilized to verify the correct execution of commands over a probably remote channel.

On an abstract level, Table 3 shows which security services are required by which use cases and which of the analyzed HTAs are fitted to address the required security service.

In UC1: Credential Management all key management operations, like key generation and derivation, exchange, storage, and usage are required to create and securely distribute credentials both during provisioning process as well as afterwards to update or revoke credentials. Depending on the specific process, attestation mechanisms may be required to first verify a benign system before key management. Also audit sessions may be relevant that give trusted evidence of executed key management operations.

In UC2: ECU Communication there is typically no key management required anymore with the possible exception of derivation or exchange of session keys. The prominent functionality is the key usage. Especially, in the communication with backend systems, vehicle attestations are important to assert the trustworthiness of the vehicle before establishing the communication. Especially for the increasing message exchange with the vehicle environment or time limited use cases like feature activation, secure time synchronization is important in UC2.

The last use case, UC3: HPC Utilization, targets all security services. The ones exclusive for UC3 are key backup, parallel access, and HWDTs. These are especially targeted at upcoming challenges

regarding the secure consolidation of vECUs on one platform and corresponding redundancy and safety concepts.

Table 3: Mapping Criteria to Use Cases

Security Service	Use Case UC1	UC2	UC3	SHE	HSM (EVITA)	TPM	HSM	TZ[1] (no FW)
Key Generation/Update	✓	(✓)	(✓)	○	●	●	(●)	(●)
Secure Storage Flexibility	✓	-	(✓)	◐	●	●	(●)	(●)
Key Usage Authorization	✓	✓	✓	○	◐	●	(●)	(●)
Key Backup	-	-	✓	○	●	●	(●)	(●)
Parallel Access	-	-	✓	○	◐	●	(●)	○
Attestation	(✓)	✓	✓	○	◐	●	(●)	(●)
Secure Timing	-	✓	✓	○	●	●	◐[2]	○
HW Watchdog Timer	-	-	✓	○	○	●	◐[2]	○
Audit	(✓)	-	✓	○	◐	●	(●)	(●)

UC1: Credential Management, UC2: ECU Communication, UC3: HPC Utilization,
[1] TrustZone, [2] Hardware feature that depends on host controller, (●): Implementation specific

As seen in Table 3, TPM, EVITA and the freely-programmable HSM as well as TrustZone have full support for key generation, e.g., based on key derivation or Diffie–Hellman (DH) key agreement protocols. Regarding key update, the TPM has a certification process where it can securely prove to remote parties the generation of new TPM-inherent keys. SHE does not support key generation but has a key update protocol to import keys securely over a remote channel. In all HTAs the keys can have usage flags that restrict usage, e.g., for authentication or encryption.

The main reason to integrate HTAs is the provision of a secure (key) storage. Thus, every analyzed HTA provides secure storage with SHE only supporting keys and the secure boot MAC. The protection guarantees vary among the analyzed HTAs (cf., Section 5.2).

Next to the key usage flags that are defined during key provisioning or update, key usage may also require authorization. SHE only has an authorization mechanism for its key update protocol, while the actual key usage does not have any authorization. The EVITA HSM supports both password and/or sealing-based authorizations. Besides the freely programmable TrustZone and HSM, the TPM offers the most flexible authorization mechanisms that may unlock both keys but also access to arbitrary data or distinct data structures in the NVM. Next to simple password-based authorizations, the TPM also allows session-based authorizations that allow to authorize keys remotely over unsecured channels. This is done either using challenge-response schemes, e.g., based on HMAC keys, or using fine-granular TPM policies. These are the most flexible mechanism where authorization can be bound to various conditions, like platform state, time or usage limitations and/or knowledge of a distinct cryptographic key. They are so flexible that even complex right and role management systems can be instantiated completely within the TPM [23].

Key backup is a service mostly used in scenarios where redundancy is important, e.g., to ensure safety requirements in autonomous driving. Next to the freely-programmable TrustZone and generic HSM, both TPM and EVITA HSM offer fine-granular backup services where keys may be restricted to be not migratable at all, only to entities with the same trust level, e.g., another EVITA HSM or TPM, or to arbitrary entities. SHE supports only to export the Random-Access Memory (RAM) key but only with the device-unique secret key. Thus, this is no backup mechanism, since it cannot be decrypted by any other entity.

Parallel access is important to utilize upcoming use cases regarding the virtualization of ECUs on one platform and concurrent access to one HTA. TPM and EVITA HSM feature parallel access with multiple sessions. However, the EVITA specification only mentions parallel access as general assumption without specific definition. SHE and TrustZone do not have possibilities for parallel access. At least the latter has some academic work for enhancing TrustZone with parallel access [17].

Regarding attestation, the TPM is designed as both RoTS and ROTR. As RoTS it can store integrity measurements securely in its PCRs and may report it securely to remote parties with its dedicated TPM2_Quote command. There exist multiple standards specifying attestation mechanisms [10, 25]. The EVITA HSM has similar working ECRs to store integrity measurements. However, a reporting command as well as attestation protocols are not specified.

Secure Timing and the HWDT are security services that need to be implemented by the hardware. Thus, they are not directly supported by TrustZone or a freely-programmable HSM. Regarding the secure timing, both TPM and EVITA HSM have a SIC that can be synchronized to Coordinated Universal Time (UTC) to enable time-based use cases, e.g., temporary vehicle access in car sharing use cases [23]. SHE lacks of such a feature. The HWDT is exclusively featured by the TPM and allows to implement safety concepts necessary for autonomous vehicles, e.g., hot-cold failover concepts [16].

Audit is necessary to verify a successful execution of commands, e.g., during remote provisioning of the HTA. SHE does not support this feature, while both TrustZone as well as the freely-programmable HSM may support this feature. The EVITA HSM has only a soft and not detailed specified requirement for auditing and audit reporting of certain important commands, command invocation failures, and internal errors. In contrast, the TPM has full audit sessions specified for various important commands.

5.2 Platform Security

The platform security evaluation criteria assesses the security of the provided shielded location regarding its tamper-resistance, e.g., physical protection, provided TCB, and the general verifiability of the specification in relation to our defined attacker model. Figure 2 is adapted from [12] and shows a comparison of the structure of the analyzed hardware trust anchor technologies with respect to their corresponding TCB.

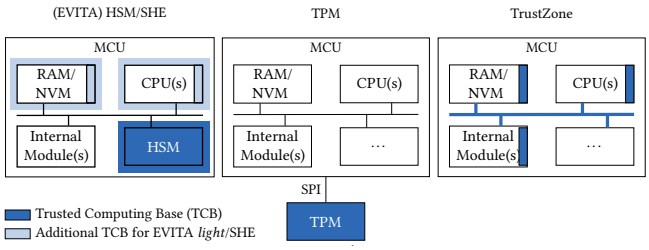


Figure 2: HTA Structure Comparison adapted from [12]

The TPM is a dedicated security co-processor that is connected to the MCU of the ECU via dedicated Serial Peripheral Interface (SPI) bus. The TCB is limited to the TPM itself since it is its own SOC

including own CPU, RAM and flash memory. It has a dedicated protection profile [34] that is conforming to the Common Criteria Version 3.1 and demands tamper resistance against physical attacks. With its tamper protection property, the TPM thus mitigates against all of our defined attacker types and offers the strongest security guarantees of all analyzed HTAs.

An HSM is a security subsystem within an SOC. With the exception of the feature sets defined by EVITA and SHE, there no specification or formal definition on the specifics on the provided security or protection level for HSMs. Every vendor may define silicon running cryptographic operations as HSM. Thus, platform security guarantees are typically unclear or even error-prone [27]. There is typically no Common Criteria (CC) certification and respective protection profile. The TCB is typically the HSM itself, but for probable tamper resistance the HSM relies on the security guarantees of the host controller where the HSM is embedded into and thus enlarges the TCB. For EVITA *light* and SHE the TCB is further enlarged to the RAM/NVM as well as to the CPU of the host controller since in this specifications they are not part of the security subsystem. While not providing tamper protection, at least EVITA *medium* and EVITA *full* mitigate against cyberattacks up to Type2 attacker since their TCB is shielded from the host controller. EVITA *light* and SHE only mitigate against Type1 attackers since their TCB also comprise parts of the probably compromised host controller.

TrustZone is an ARM processor extension that initiates a TEE for security-sensitive applications that runs in parallel on the host controller. The TCB is thus the TEE-part of CPU, RAM/NVM, internal modules and peripherals as well as the connecting bus system. It provides no tamper-protection and there were found multiple critical attacks on TrustZone for multiple manufacturers [2]. Since by design its TCB utilizes all components of the host controller, privilege escalation attacks may compromise the security guarantees and thus it only mitigates against Type1 attackers.

5.3 Software Ecosystem

Figure 3 shows the software ecosystem of the analyzed HTAs.

The TPM has a full-fledged, free and open source ecosystem [3] consisting of the TPM Software Stack, driver, and various application software, e.g., OpenSSL or PKCS11 integration as well as command line tools for rapid prototyping. Typical Linux distributions already include TPM software into their package repositories.

Software for EVITA HSM and SHE is typically only commercially available and not openly accessible for evaluation. Some related work of automotive Tier 1 suppliers with access to the corresponding HTAs [27] analyzed the readiness of HSM software and found out that advertised software is often not production ready since it may not be fully implemented, usable, or tested.

For the generic freely-programmable HSM there exists typically only proprietary drivers to flash own software. Thus, no other software is available and must be developed additionally.

Regarding TrustZone, the Trusted Firmware Project provides some reference open source firmware and software implementations [19], e.g., an open source TEE (OP-TEE) that leverages the Arm TrustZone technology. However, specific trusted applications still need to be implemented for the TEE. There exist also some proprietary TEE implementations from the mobile device world, e.g., SierraTEE [31] or Samsung TEEGRIS [29], but they are specifically focused towards mobile use cases and thus out of scope here.

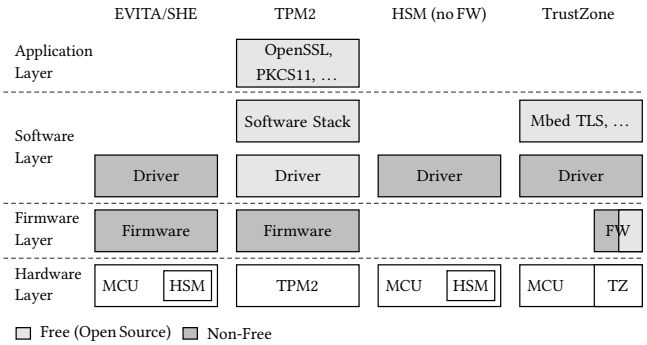


Figure 3: HTA Software Module Comparison

5.4 Cryptographic Performance

Fast cryptographic processing of the HTAs is crucial for use cases that need high-performance cryptographic processing, e.g., generation and verification of digital signatures in V2X use cases [1].

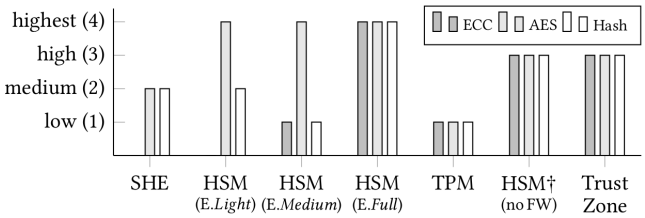
However as stated in the background section, there is a variety of different implementations for the analyzed HTAs that slightly differ in performance properties. Thus, we make only an ordinal performance comparison here that relates the performance of each HTA among each other. We introduce the performance rating score 1 (low), 2 (medium), 3 (high), and 4 (highest). A HTA rated 1 or 2 has no dedicated cryptographic accelerators but a HTA rated with 2 can at least utilize the ECU CPU that is typically faster then the embedded CPU of a 1-rated HTA. HTA rated with 3 typically has cryptographic accelerators or can utilize a fast application CPU and a 4-rated HTA is specified to have cryptographic accelerators. Figure 4 summarizes our results that are detailed in the following.

The only HTAs featuring dedicated cryptographic hardware accelerators are the EVITA HSMs, so that they are faster than any other of the analyzed HTAs and obtain a rating of 4. EVITA *full* has cryptographic hardware accelerators for ECC, AES, and the Whirlpool hash function, EVITA *medium* and EVITA *light* have only AES cryptographic hardware accelerators. EVITA *light* has no support for ECC but can utilize the typically faster host processor for hash operations.

Slightly slower with a rating of 3 are the freely-programmable HSM and TrustZone. For the freely-programmable HSM we just assume an average performance since this depends on the specific implementation, e.g., if and which cryptographic hardware accelerators are implemented. TrustZone has typically no dedicated cryptographic accelerator but by design utilizes the fast host processor and thus also gets the same performance rating.

SHE does not have any cryptographic hardware accelerators but also utilizes the host CPU of an ECU. The ECU CPU is typically slower than the application processor of TrustZone-enabled MCU and thus SHE is rated with 2.

Regarding performance, the TPM is ranked with 1 since it neither has cryptographic hardware accelerators nor utilizes a fast application or ECU CPU.



† Depends on specific implementation. Chosen Average.

Figure 4: Ordinal Cryptographic Performance Comparison of the selected HTAs

5.5 Flexibility

Both TrustZone and the generic HSM are freely-programmable. Thus, they are the most flexible of the analyzed HTAs regarding non-hardware dependent supported features.

All other analyzed HTAs have a fixed feature set and thus are not flexible regarding the provided security services. An exception may be TPM. Its specification is designed crypto-agile so that it is already prepared for the addition of new cryptographic algorithms and primitives. This is shown in the cryptographic registry which is constantly enhanced with new cryptographic algorithms and primitives [33]. In addition, research is underway to make the TPM specification ready for post-quantum cryptography [6, 21].

5.6 Cost Efficiency

The cost efficiency of a HTA is influenced by the available software ecosystem. Figure 3 shows that the software for the TPM is freely available and open source, so that next to the hardware costs of the TPM chip there is no additional software development costs.

All other HTAs have the benefit that the security subsystem or TEE is already present on the controller and thus the overall hardware acquisition costs are typically cheaper than a dedicated hardware security chip like the TPM. However, our software ecosystem analysis (cf. Section 5.3) has shown that these HTAs come with additional software costs to license the non-free software components or even develop custom software. Moreover, our analysis of related work [27] has shown, that advertised software is often not production-ready and additional costs may be introduced to further tailor the software to the needs of the customer.

6 RESULT SUMMARY AND DISCUSSION

Table 4 summarizes our evaluation and introduces a score ranging from 0 (criterion not met) to 4 (criterion fully met) to assess the degree to which our evaluation criteria are met.

Regarding the provided security services (EC1), the HTAs with freely-programmable firmware, like HSM without firmware and TrustZone, are proven to be most flexible and thus obtain the highest rating of 4. With 3, the TPM is slightly lower rated, since despite its fixed feature set it supports all required security services from our use cases (Table 2) and has a agile specification that is designed to be extendable. The EVITA *full* specification obtains a rating of 2, The EVITA *medium* a rating of 1, and both EVITA *light* and SHE a rating of 0 according to the decrease in provided functionality.

Table 4: Hardware Trust Anchor Evaluation Summary.

Technology	EC1: Security Services	EC2: Platform Security	EC3: Flexibility	EC4: Performance	EC5: Software Ecosystem	EC6: Cost Efficiency	Sum
HSM (no FW)	4	2†	4	2†	0	0	12
SHE	0	1	0	1	1	1	4
EVITA *full*	2	2	0	4	1	2	11
EVITA *medium*	1	2	0	4	1	2	10
EVITA *light*	0	1	0	4	1	2	8
TPM	3	4	1	0	3	3	14
TrustZone	4	0	4	2†	1	1	12

Evaluation scale from 0 (criterion not met) to 4 (criterion fully met).
† Depends on specific implementation. Chosen Average.

Regarding the provided platform security guarantees (EC2), the TPM obtains the highest rating of 4, since it is a dedicated security chip with the smallest TCB. Moreover, it has tamper-protection and a open and well-evaluated specification. Thus, it mitigates against all of our defined attacker types. All other HTAs obtain a lower rating since platform security properties are not (openly) specified and trust in the controller manufacturer is necessary. At least, EVITA *full* and EVITA *medium* have a dedicated CPU and RAM and thus mitigates against up to Type2 attackers. EVITA *light* and SHE share resources with the host controller extending the TCB even further so that they mitigate only against Type1 attackers. TrustZone has the biggest TCB since by design all components of the host controller are shared with the secure world of TrustZone, which results in a rating of 0. Still it protects against Type1 attackers.

Also regarding flexibility (EC3), the freely-programmable HTAs, like HSM without firmware and TrustZone, score the highest rating of 4 since they can be specifically tailored to the system requirements. All other HTAs have a fixed feature set and thus are rated with the 0. An exception is the TPM, where at least the specification is designed to be extensible. While the HSM feature sets SHE and EVITA are sometimes further enhanced by manufacturers, e.g., SHE+, with new cryptographic algorithms and primitives, the corresponding specification details are, if existing, not publicly available. Thus, the underlying security guarantees or deviation from the original specification are not verifiable.

The HTAs with cryptographic accelerators obtain the highest performance ratings. These are EVITA *full*, EVITA *medium*, and EVITA *light* with a rating of 4. Both the freely-programmable HSM and TrustZone obtain a rating of 2, since the real performance depends on the implementation and thus we assume a medium value here. SHE has no cryptographic accelerators but at least uses the typically faster host controller CPU and thus scores 1. The TPM has no accelerators and uses its own typically slow CPU. From all analyzed HTAs it is the slowest regarding performance. However, there are possibilities of a TPM-backed software processing with an obvious security tradeoff.

Regarding the software ecosystem, the TPM offers a broad range of free and open source software that comprises firmware, driver, software stack as well as application software. For the HSM and

SHE, available firmware is mostly proprietary and then only commercially available. For TrustZone exists open source firmware and software reference implementations from ARM, but commercially available solutions are mostly bundled with commercially, non-free firmware and additional software also needs to be implemented.

7 CONCLUSION

In this work, we have analyzed and evaluated HTAs regarding their feasibility to secure modern and upcoming automotive architectures. We derived relevant evaluation criteria from both related work as well as an extensive automotive domain analysis where we defined a future reference architecture, an attacker model, as well as current and future automotive use cases. For the following evaluation we make an extensive analysis of the corresponding specifications and map our evaluation criteria to the results of our automotive domain analysis.

Our evaluation has shown relevant advantages and disadvantages of the analyzed HTAs. In particular, the results show that the currently predominantly used HSMs are feasible for modern use cases but lack of tamper protection guarantees for physical attackers and certain features for upcoming use cases. While not yet widespread for automotive use cases, the TPM has proven to be versatile and can be mapped to both the EVITA and SHE feature sets. In addition, it has some advanced hardware-backed features that cannot be emulated by software in freely-programmable HTAs.

ACKNOWLEDGMENTS

This research work has been partly funded by the German Federal Ministry of Education and Research (BMBF) and the Hessen State Ministry for Higher Education, Research and the Arts within their joint support of the National Research Center for Applied Cybersecurity ATHENE and by the BMBF project SAVE (ID 16KIS1324). Additionally, the project leading to this application has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 883135 (E-Corridor).

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